

Jiale Zhao

Research Assistant, Tsinghua University

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Personal Profile

I am a research assistant at Tsinghua University, working on AI for Science at the intersection of genomics and large language models (LLMs), with a focus on studying gene evolution. I earned a B.Eng. in Computer Science from Chongqing Univ. of Posts and Telecommunications in 2026 and will begin a CS Ph.D. at Penn State University (PSU) in Spring 2027. Previously, I was an LLM algorithm intern at Li Auto. I also work on co-evolving documentation for coding agents (CoEvoDoc) and behavior-oriented persona learning with Prof. Lu Cheng (PSU).

My interests include **AI for Science and genomics**, **human-centered HCI**, **agentic/multi-step reasoning**, **interpretability/controllability**, **self-evolving systems**, and **multimodal pipelines**.

Education

Incoming Ph.D., Computer Science

Penn State University

University Park, PA

Starting Spring 2027

B.Eng., Computer Science

Chongqing Univ. of Posts and Telecommunications

Chongqing, China

2021 – 2026

Publications

- **Jiale Zhao**, Ke Fang, and Lu Cheng. **When and What to Ask: AskBench and Rubric-Guided RLVR for LLM Clarification** (Findings of ACL 2026; arXiv: <https://arxiv.org/abs/2602.11199>)
- Sunzhu Li, **Jiale Zhao**, Miteto Wei, Huimin Ren, Yang Zhou, Jingwen Yang, Shunyu Liu, Kaike Zhang, and Wei Chen. **RubricHub: A Comprehensive and Highly Discriminative Rubric Dataset via Automated Coarse-to-Fine Generation** (ACL 2026 main; arXiv: <https://arxiv.org/abs/2601.08430>)
- Sunzhu Li, Zhiyu Lin, **Jiale Zhao**, Shuling Yang, and Chen Wei. **ThinkPilot: Steering Reasoning Models via Automated Think-prefixes Optimization** (Findings of EACL 2026; paper: <https://aclanthology.org/2026.findings-eacl.185/>)
- Zhiyu Lin, Jingwen Yang, **Jiale Zhao**, Meng Liu, Sunzhu Li, and Benyou Wang. **Decoding the Ear: A Framework for Objectifying Expressiveness from Human Preference Through Efficient Alignment** (Interspeech 2026; Oral; arXiv: <https://arxiv.org/abs/2510.20513>)
- Yang Zhou, Sunzhu Li, Shunyu Liu, Wenkai Fang, Kongcheng Zhang, **Jiale Zhao**, et al. **Breaking the Exploration Bottleneck: Rubric-Scaffolded Reinforcement Learning for General LLM Reasoning** (ICML 2026 main; arXiv: <https://arxiv.org/abs/2508.16949>)

Under Review

- **Beyond Outcome Rubrics: Process Rewards for Non-Verifiable Tasks** (Under review)
- **Not All Correct Responses Are Equal: Behavior Matters in Synthetic Data Generation for Post-Training** (Under review)

Ongoing Work

- **CoEvoDoc** (first; collaboration with Prof. Lu Cheng, PSU) — a co-evolving repository documentation system for coding agents that adapts document structure complexity based on project change magnitude, maintaining higher localization stability, modification correctness, and doc-code consistency during long-term repo evolution.

Research Assistant

- **AI for Science** — investigating gene evolution by combining genomics and large language models.

Sep 2026 – present

LLM Algorithm Intern

Li Auto, Beijing

- **Deep Search** — built high-quality search trajectories on a customized Codex framework to boost agent deep search capabilities.
- **Data Flywheel for Code LLM** — developed an evaluation-centric loop (SFT → evaluate → data build → filtering → back to SFT) to raise coding capabilities.
 - Evaluation-first: standardized harnesses and rubrics so harder tasks surface real capability instead of being bottlenecked by today's noisy, underspecified eval.
 - Linked loops: evaluation feedback drove tougher data construction; the same tooling filtered/ repaired low-quality samples before feeding them back into SFT.
- **Multi-step Reasoning + Tool Invocation Agent** — developed a code-LLM agent that plans, writes code, and executes tool calls for precise answers.
 - Multi-step reasoning: decomposed complex or debugging asks into structured plans so context could be stitched into a single executable query.
 - Tool grounding: integrated code execution and API/function calls for live data or environment actions when pretrained knowledge fell short.
- **MindGPTo (GPT-4o-style multimodal app)** — end-to-end audio + vision system with paralinguistic control, built from scratch with modular FE/BE interfaces.
 - Mode coverage: supported audio→ASR→LLM→TTS, audio2text→TTS, audio2audio, and multimodal audio+image+video→text→TTS workflows.
 - Paralinguistic SFT: audio data pipelines expanded beyond laughter/pause to age, gender, compound emotions, emotional actions, and ambient sounds.

Sep 2023 – Jul 2026